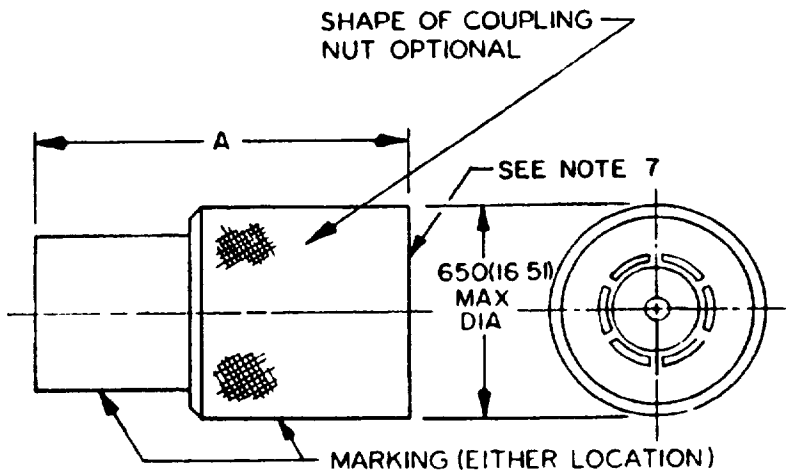
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUGS, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES QNC, (CABLED, PIN CONTACT, CLASS 2)

INACTIVE FOR NEW DESIGN AFTER 3 October 1986

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE III

NOTES:
1. Dimensions are in inches.
2. For dimensions A see table I and II.
3. Dimension .650 is the largest overall diameter of the connector.
4. Metric equivalents are given for general information only.
5. Wrench flats to accommodate standard wrench in accordance with FED-STD-H28, appendix 10.
6. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
7. Series QNC, pin contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

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TABLE I. Dash numbers, usage cross-reference, and basic overall dimensions.

Dash no. 1/ 4/ 5/	Applicable cable no. =	Dimensions	Inches-millimeters maximum 2/
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED)			
0001	M17/28-RG058 M17/155-00001ø M17/84-RG223* M17/167-00001ø M17/60-RG142ø M17/158-00001ø	A	1.250 (31.75)
0002	M17/29-RG059*Δ M17/30-RG062Δ M17/90-RG071Δ M17/110-RG302øΔ M17/97-RG210Δ	A	1.250 (31.75)
0003	M17/54-RG122* M17/157-00001ø	A	1.250 (31.75)
0004	M17/111-RG303ø* M17/170-00001ø	A	1.250 (31.75)
CATEGORY C - FIELD REPLACEABLE (MIL-C-22520/5-01 BASIC CRIMP TOOL) 3/			
0010	M17/28-RG058*~ M17/155-00001~ø M17/111-RG303ø~ M17/170-00001~ø	A	1.500 (38.10)
0011	M17/84-RG223*~ M17/167-00001~ø M17/60-RG142ø~ M17/158-00001~ø	A	1.500 (38.10)
0012	M17/29-RG059*Δ> M17/30-RG062Δ> M17/110-RG302øΔ> M17/97-RG210Δ>	A	1.500 (38.10)
0013	M17/54-RG122*& M17/157-00001&ø	A	1.500 (38.10)
0014	M17/90-RG071*Δ>	A	1.500 (38.10)

1/ Typical mating connectors for each dash number are those connectors of MIL-C-39012/66 which are of the same category and utilize the same cable.

2/ Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm. Millimeters are in parentheses.

3/ Category C connectors are assembled by means of the applicable crimping tool per MIL-C-22520 to the specified cable stripping in accordance with figure 2.

4/ These connectors shall be furnished with noncaptive center contacts.

5/ Inactive for new design. Not for Air Force or Navy use.

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≡ The RG cables are specified with the basic number. The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes ø and Δ.

ø Cable to be used for the +200°C thermal shock tests.

Δ These are not 50 ohm cables, therefore, when attached to specified connectors, VSWR, RF leakage and insertion loss are not applicable.

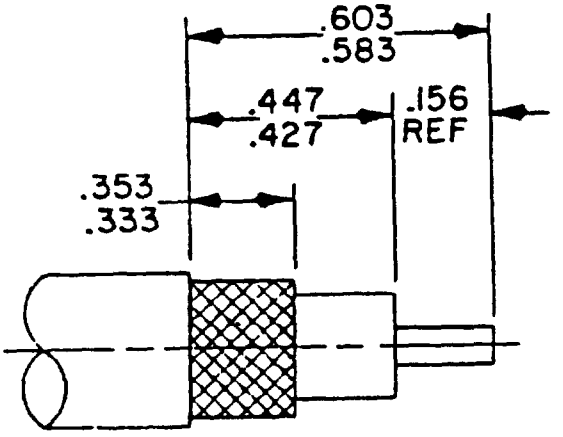
& M22520/5-41 Closure B or M22520/5-05 Closure B
M22520/5-09 Closure A

~ M22520/5-19 Closure B or M22520/5-05 Closure A
M22520/5-11 Closure A
M22520/5-57 Closure A

> M22520/5-19 Closure A or M22520/5-07 Closure A
M22520/5-13 Closure A
M22520/5-59 Closure A

ø Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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Inches	mm
.156	3.96
.333	8.46
.353	8.97
.427	10.85
.447	11.35
.583	14.81
.603	15.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Cable stripping dimensions for category C connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

500 volts rms maximum working voltage.

125 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque - 2 inch-pounds maximum.

Coupling proof torque: 20 inch-pounds minimum.

Inspection conditions: 4 to 6 inch-pounds.

Mating characteristics:

See MIL-STD-348.

Outer contact:

Test ring ID - .319 maximum, 16 microinches finish.

Insertion force: 5 pounds maximum when inserted a minimum of .093.

Contacts with slotted members: Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: 6 pounds minimum.

Resistance to test prod damage: Not applicable.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower. 1.30 maximum.

Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.008 + .002 F (F in GHz).

Item 16 - VSWR shall be less than 1.008 + .002 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.012 + .004 F (F in GHz).

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Group B inspection - VSWR shall be less than 1.024 +.007 F (F in GHz).

Group C inspection - VSWR shall not exceed 1.08.

Connector durability:

Insertion and withdrawal force: 500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact:	2.0	2.5
Outer contact:	.2	Not applicable
Braid to body:	.1	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 2,000 volts rms at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B.

Shock: Method 213 of MIL-STD-202, test condition I.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and II).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude - 70,000 feet.

Voltage - 375 volts.

RF high potential withstanding voltage:

Voltage and frequency - 1,000 volts rms at 5 MHz.

Leakage current - Not applicable.

Cable retention force:

Non-crimp assemblies. 75 pounds minimum.

Crimp assemblies:

50 pounds minimum for cables .155-.189 OD.
60 pounds minimum for cables .190-.229 OD.
75 pounds minimum for cables .230-.249 OD.
90 pounds minimum for cables .250-and larger.

Coupling mechanism retention force: 100 pounds minimum.

RF leakage: -60 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss. .2 dB maximum at 10 GHz.

Qualification not required.

Part number: M39012/65 (dash number from table I or "B" number from table II).

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TABLE II. Category B-nonfield replaceable (special tools may be required).

NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY

Dash no. 1/ 3/ 4/	Applicable cable no. ≠	Dimensions	Inches-millimeters maximum 2/
CATEGORY B - NONFIELD REPLACEABLE (SPECIAL TOOLS MAY BE REQUIRED) 4/			
0005	M17/28-RG058* M17/155-00001ϕ	A	1.500 (38.10)
0006	M17/84-RG223* M17/167-00001ϕ	A	1.500 (38.10)
0007	M17/29-RG059*Δ M17/30-RG062Δ M17/110-RG302@Δ M17/97-RG210Δ	A	1.500 (38.10)
0008	M17/54-RG122* M17/157-00001ϕ	A	1.500 (38.10)
0009	M17/90-RG071*Δ	A	1.500 (38.10)
0015	M17/111-RG303@* M17/170-00001ϕ	A	1.500 (38.10)
0016	M17/60-RG142@* M17/158-00001ϕ	A	1.500 (38.10)

1/ Typical mating connectors for each dash number are those connectors of MIL-C-39012/66 which are of the same category and utilize the same cable.

2/ Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm. Millimeters are in parentheses.

3/ These connectors shall be furnished with noncaptive center contacts.

4/ Inactive for new design. Not for Air Force or Navy use.

≠ The RG cables are specified with the basic number. The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C thermal shock tests.

Δ These are not 50 ohm cables, therefore, when attached to specified connectors, VSWR, RF leakage and insertion loss are not applicable.

ϕ Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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TABLE III. Cross-reference of part numbers.

Part number 1/ M39012/65B	Superseded part number M39012/65-
0005	0005
0006	0006
0007	0007
0008	0008
0009	0009
0015	0015
0016	0016

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI, EA
Navy - SH
Air Force - 11, 17, 80, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3518-24)

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